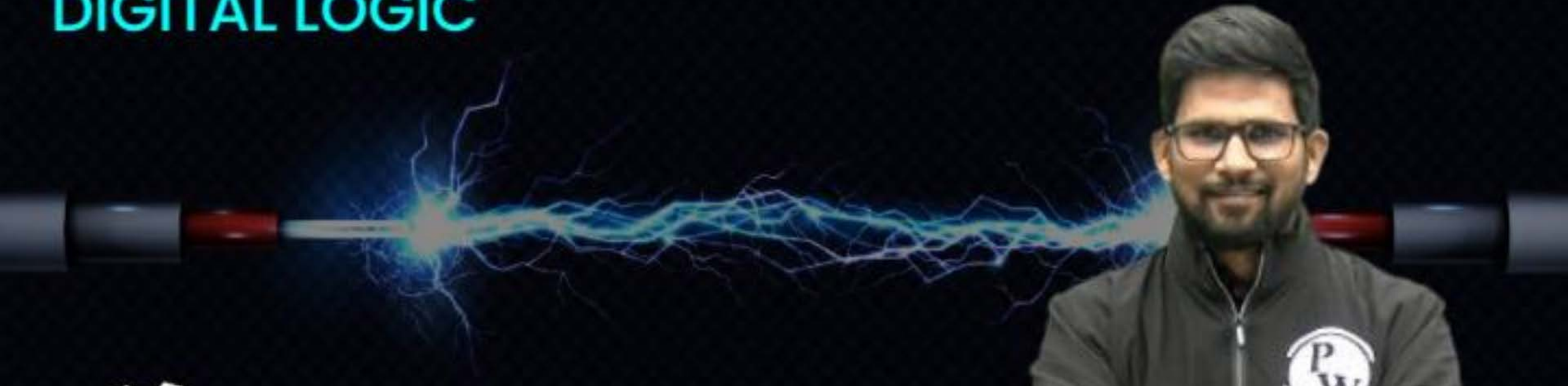
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 04

BOOLEAN THEOREMS AND GATES



By- Chandan Gupta Sir



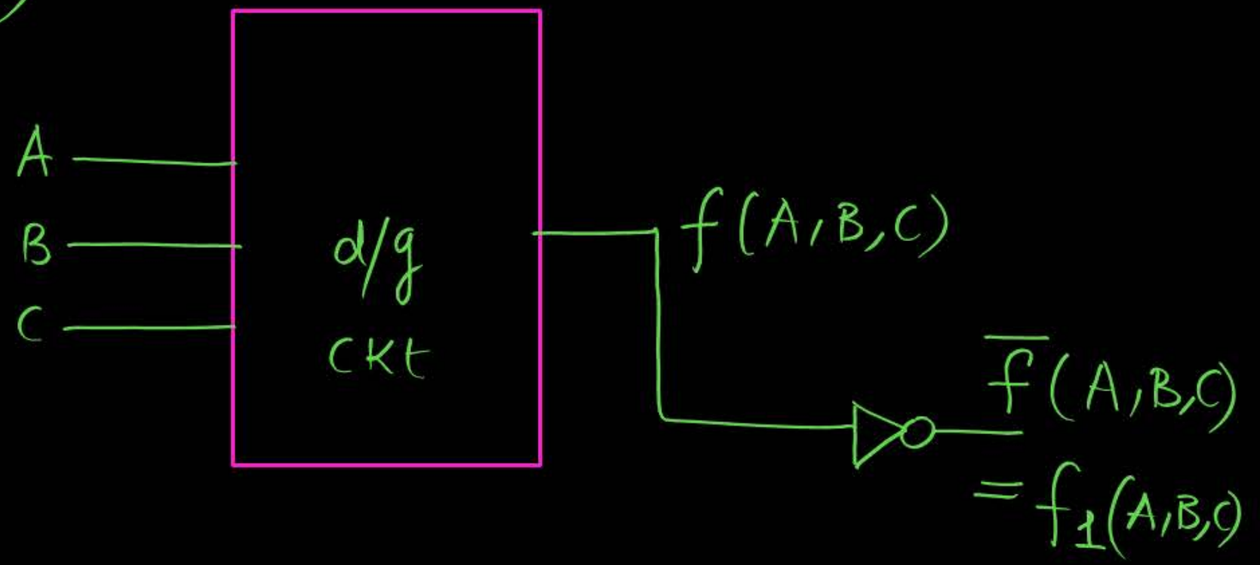
Theorems & some imp basics:



Logic gates.

- $f(A,B,C) = \Sigma(1,2,6,7) = \pi(0,3,4,5)$

$\overline{f}(A,B,C) = f_1(A,B,C) = \Sigma(0,3,4,5) = \pi(1,2,6,7)$



- $f(A,B,C) = \Sigma(0,2,3)$

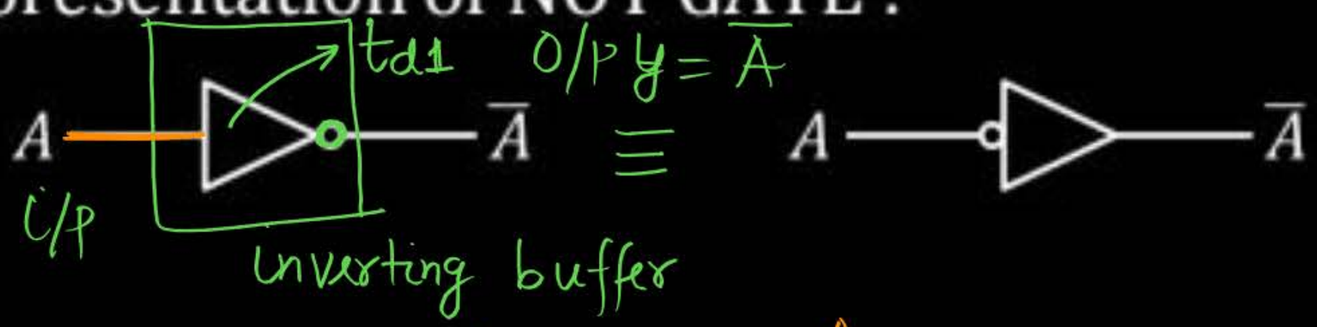
$\overline{f}(A,B,C) = \Sigma(1,4,5,6,7)$

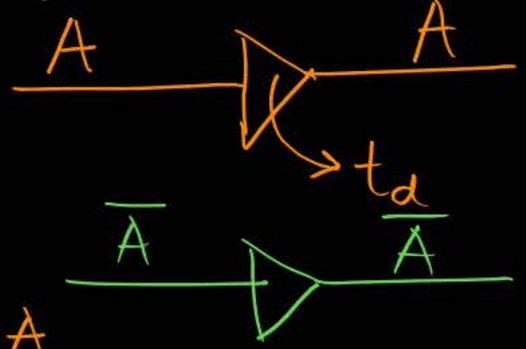
GATES

- Basic Gates : NOT, AND, OR
- *** Arithmetic Gates : XOR & XNOR
- // Universal Gates : NAND & NOR

[NOT GATE]

Representation of NOT GATE :



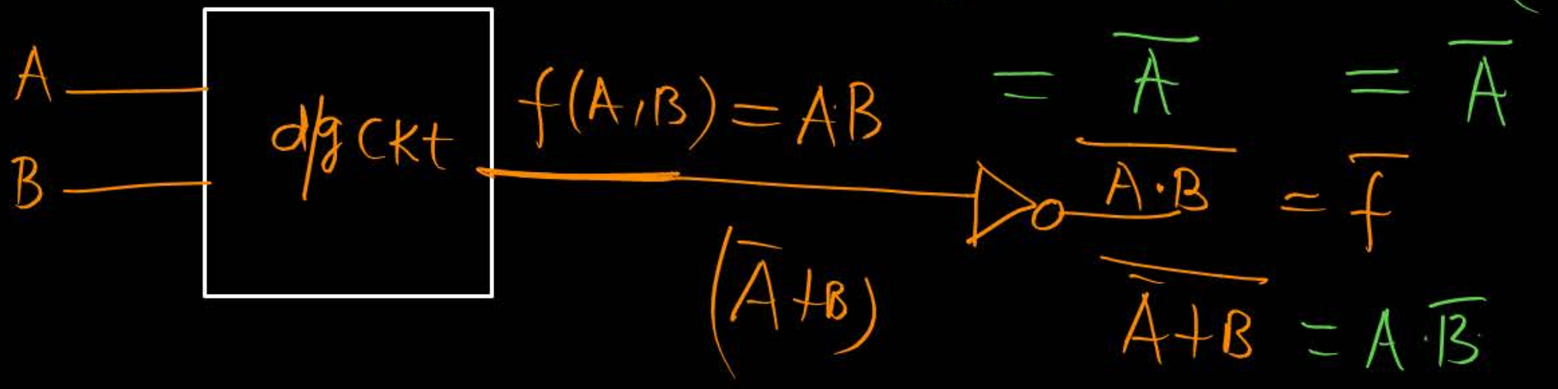


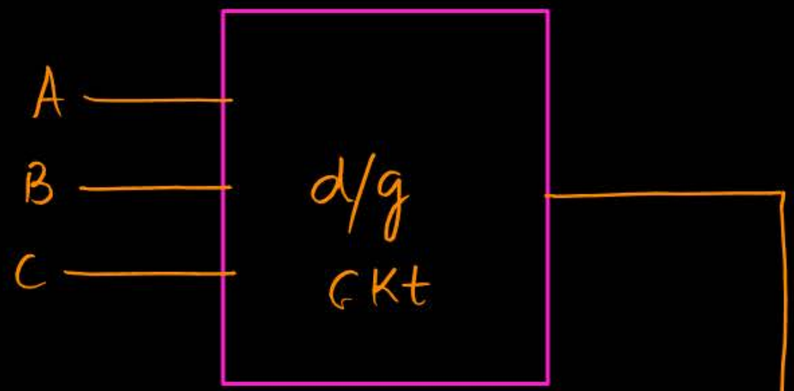
Input A	Output Y
0	1
1	0

'0'

'1'

$y(A) = \Sigma(0) = \Pi(1)$





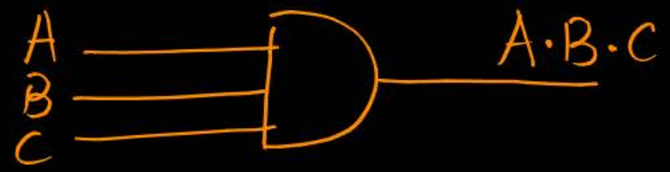
$f(A,B,C) = (AB + \bar{B}C)$

$$\bar{f} = \overline{AB + \bar{B}C}$$

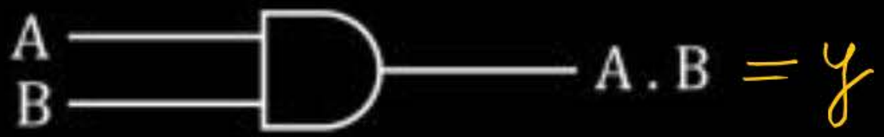
$$= \overline{AB} \cdot \overline{\bar{B}C}$$

$$= (\bar{A} + \bar{B})(B + \bar{C})$$

[AND GATE]



Representation :



	2^1	2^0	
	A	B	Output y = A.B
0	0	0	0
1	0	1	0
2	1	0	0
3	1	1	1

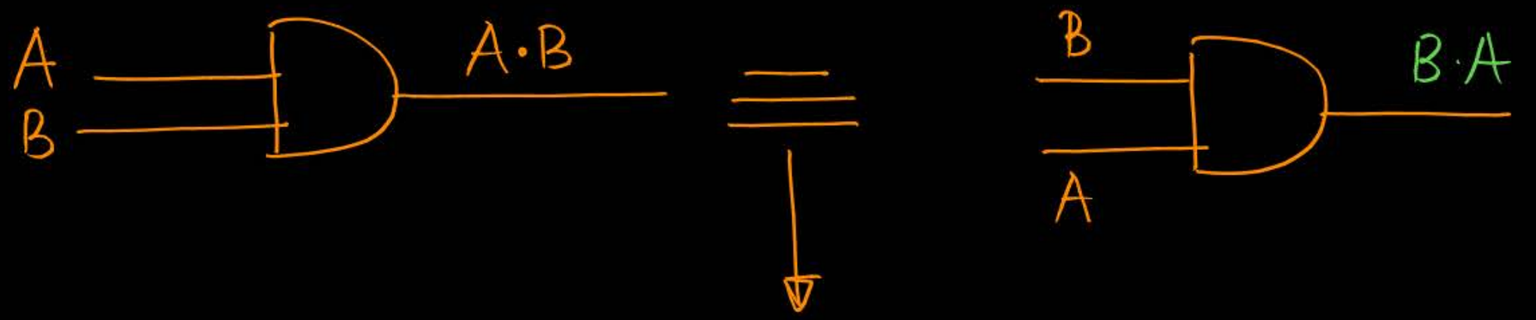
$y(A,B) = \Sigma(3) = \pi(0,1,2)$

$= A \cdot B = (A+B) \cdot (A+\bar{B}) \cdot (\bar{A}+B)$

$= \left[A + B\cdot\bar{B}\right]\left[\bar{A}+B\right]$

$= A\cdot\bar{A} + A\cdot B = AB$

- Commutative Law :

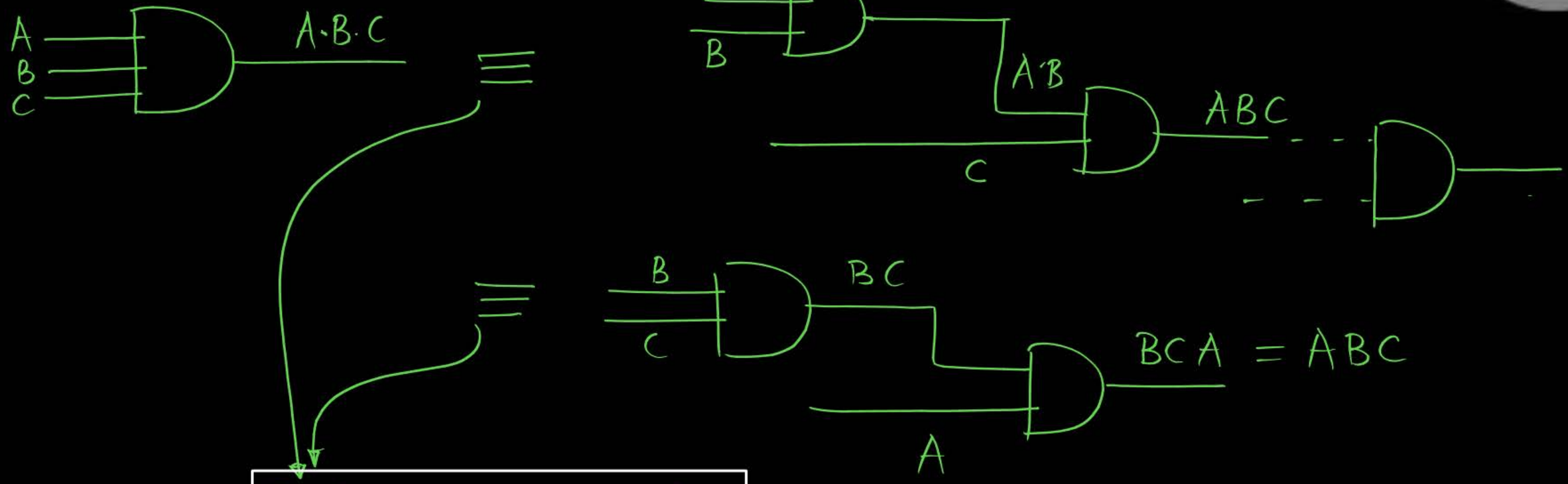


$A \cdot B = B \cdot A$

It holds commutative law.

Position of variables is irrelevant.

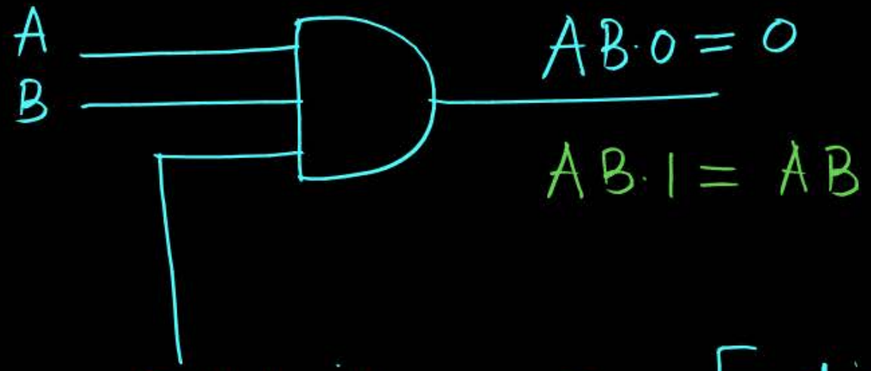
- Associative Law :



It holds associative law.

→ this signifieses that we can generate multi i/p AND operation using 2-i/p AND operation.

- Enable and Disable input for AND gate :



Control i/p = 0 [disable i/p for AND gate]

= 1 [enable i/p for AND gate]

- IMP points :

- If any one of the i/p line is at logic '0' then irrespective of other i/p lines o/p will definitely be '0'.

- O/p will be '1' only if all the i/p lines are '1' together.

[OR GATE]



Representation :



	2^1	2^0	
	A	B	$y = A + B$
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	1

$y(A,B) = \Sigma(1,2,3) = \pi(0) = (A+B)$

$= \bar{A}B + A\cdot\bar{B} + AB$

$= A + (\bar{A}\cdot B)$

$= (A+\bar{A})(A+B)$

$= (A+B)$

- Commutative Law :

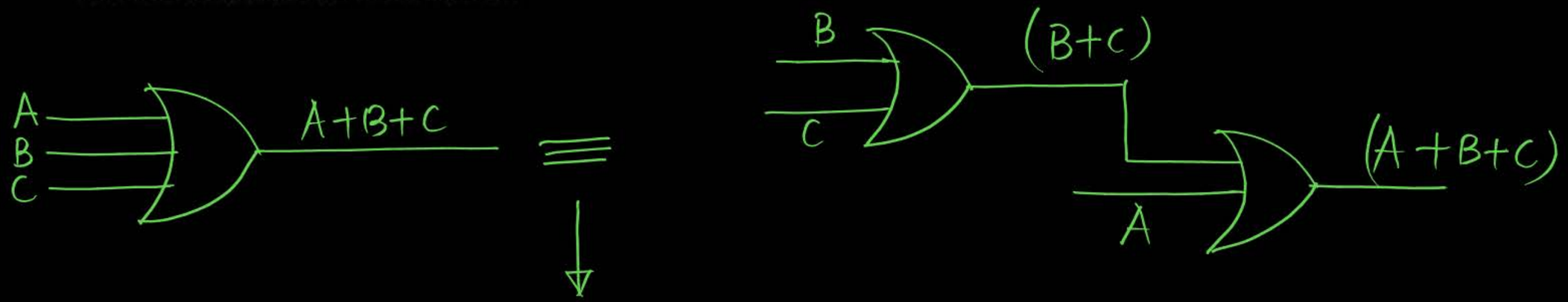


$(A+B)=(B+A)$

It holds commutative law.

position of variables is irrelevent.

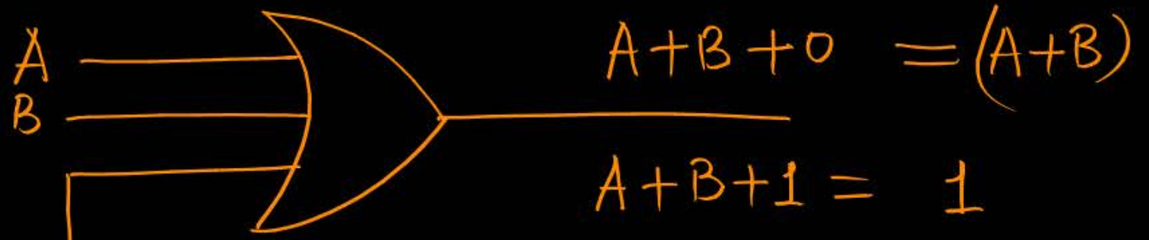
- Associative Law :



It holds associative law.

→ multi i/p OR operation can be generated using 2-i/p OR operation.

- Enable and disable input for 'OR gate' :



Control i/p = 0 [enable i/p for OR gate]

= 1 [disable i/p for OR gate]

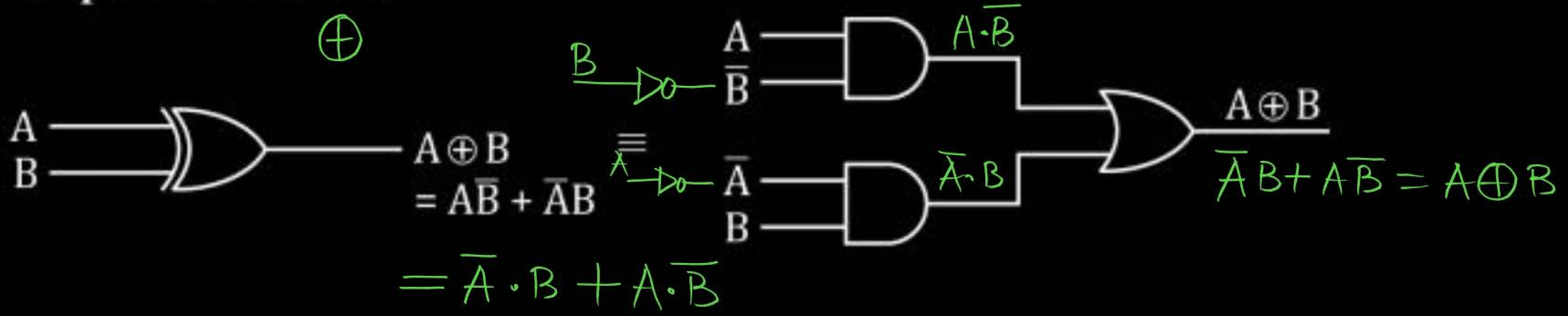
- IMP Points :

- If any one of i/p line is at logic '1' then irrespective of other i/p lines o/p will definitely be '1'.
- o/p will be '0' only if all the i/p lines are '0' together.

[XOR GATE]

'Exclusive OR' ⟶ It is defined for two i/Ps only.

Representation :



A	B	$y = A \oplus B$
0	0	
0	1	
1	0	
1	1	

$y(A,B) =$

- $A \oplus B = \overline{A} \cdot B + A \cdot \overline{B}$
- $AB \oplus BC = \overline{AB} \cdot BC + AB \cdot \overline{BC}$
- $A \oplus (A+B) = \overline{A} \cdot (A+B) + A \cdot \overline{(A+B)}$
- $A\overline{B} \oplus CD = \overline{A\overline{B}} \cdot CD + (A\overline{B}) \cdot \overline{CD}$

H·W·

- Q.1: $f(A,B,C) = \Pi(0,1,2,3,4)$

  $\overline{f}(A,B,C) = \Pi(\ ?\ ) = \Sigma(?)$

Q.2 $f(A,B,C) = \Sigma(0,2,4,6)$

$\overline{f(A,B,C)} = \Pi(\ ?\ )$

Q.3 $f(A,B,C,D) = \Sigma(0,2,4,6,7,9,10,11,14)$

$\overline{f(A,B,C)} = \Sigma(?)$



2 Minute Summary

NOT, AND, OR , XOR

Thank you

GW

Soldiers !

